Unit

16 Salts and neutralization

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Acid-base neutralization
- Preparation of salts
- Uses of neutralization

16.1 Acid-base reactions

When the right amounts of an acid and a base are mixed, they react completely to produce a salt and water only.

The reaction between an acid and a base is called neutralization. N1 We can divide acid-base neutralization reactions into three main types as listed below.

Neutralization of an acid and an alkali

An acid reacts with an alkali to produce a salt and water.

✓ acid + alkali ⟶ salt + water

For example, dilute hydrochloric acid reacts with dilute sodium hydroxide solution to give sodium chloride and water.

$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

We can use litmus paper to test whether the resulting salt solution is acidic or alkaline (Fig. 16.1).

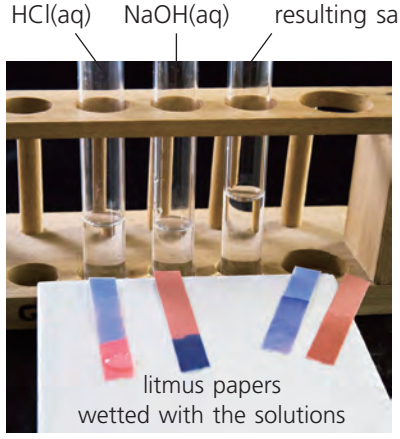


Fig. 16.1 When the right amounts of HCl(aq) and NaOH(aq) are mixed, they react completely to give NaCl and water only. Litmus papers show that the resulting salt solution is neither acidic nor alkaline

N1 Pairs of substances undergoing chemical reactions when mixed
Examination questions often give pairs of substances and ask student to identify pairs which undergo a chemical reaction when mixed.

Examples:

- vinegar and antacid
- magnesium oxide and dilute ethanoic acid
- sodium hydroxide solution and dilute ethanoic acid

} undergo a chemical reaction when mixed

- bleach and antacid
- vinegar and table salt
- copper and dilute ethanoic acid

} do NOT undergo a chemical reaction when mixed

Reference Website
Refer to the video clip 'Neutralization':
http://resources.hkedcity.net/resource_detail.php?rid=1065476039
(accessed July 2014)

Since HCl, NaOH and NaCl dissociate completely in water, we can write these compounds in ionic forms:

$$H^+(aq) + \cancel{Cl^-(aq)} + \cancel{Na^+(aq)} + OH^-(aq) \longrightarrow \cancel{Na^+(aq)} + \cancel{Cl^-(aq)} + H_2O(l)$$

Notice that $Na^+(aq)$ and $Cl^-(aq)$ are present on both sides of the equation. When we delete ions common to both sides of the equation, we get the net ionic equation for the reaction between HCl(aq) and NaOH(aq):

$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$$

Thus, the only chemical change that occurs is the reaction between $H^+(aq)$ ions and $OH^-(aq)$ ions to give water.

Neutralization of an acid and an insoluble metal hydroxide

An acid reacts with an insoluble metal hydroxide to produce a salt and water.

CE 2005 Paper 2 Q38

✓ acid + insoluble metal hydroxide ⟶ salt + water

For example, iron(III) hydroxide is an insoluble metal hydroxide. It reacts with dilute nitric acid to give iron(III) nitrate and water.

$$3HNO_3(aq) + Fe(OH)_3(s) \longrightarrow Fe(NO_3)_3(aq) + 3H_2O(l)$$

or $$3H^+(aq) + Fe(OH)_3(s) \longrightarrow Fe^{3+}(aq) + 3H_2O(l)$$

Neutralization of an acid and an insoluble metal oxide

An acid reacts with an insoluble metal oxide to give a salt and water.

CE 2005 Paper 2 Q38

✓ acid + insoluble metal oxide ⟶ salt + water

For example, copper(II) oxide is an insoluble metal oxide. It reacts with dilute sulphuric acid to give copper(II) sulphate and water.

Practice 16.1

1 a) $2HCl(aq) + CaO(s) \longrightarrow CaCl_2(aq) + H_2O(l)$
$2H^+(aq) + CaO(s) \longrightarrow Ca^{2+}(aq) + H_2O(l)$
b) $H_2SO_4(aq) + 2KOH(aq) \longrightarrow K_2SO_4(aq) + 2H_2O(l)$
$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$
c) $2HNO_3(aq) + Pb(OH)_2(s) \longrightarrow Pb(NO_3)_2(aq) + 2H_2O(l)$
$2H^+(aq) + Pb(OH)_2(s) \longrightarrow Pb^{2+}(aq) + 2H_2O(l)$

2 a) $ZnCO_3(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2O(l) + CO_2(g)$
b) No.
In neutralization, salt and water are the only products.

$H_2SO_4(aq) + CuO(s) \longrightarrow CuSO_4(aq) + H_2O(l)$

or $2H^+(aq) + CuO(s) \longrightarrow Cu^{2+}(aq) + H_2O(l)$

We can define neutralization as follows:

✓ Neutralization is the combination of hydrogen ions and hydroxide ions (or oxide ions) to form water molecules.

In neutralization, salt and water are the only products.

Practice 16.1

1 Write a chemical equation and an ionic equation for each of the following neutralization reactions:

a) Dilute hydrochloric acid and solid calcium oxide

b) Dilute sulphuric acid and dilute potassium hydroxide solution

c) Dilute nitric acid and solid lead(II) hydroxide

2 Consider the reaction between solid zinc carbonate and dilute sulphuric acid.

a) Write a chemical equation for the reaction.

b) Is the reaction a neutralization reaction? Explain your answer.

16.2 Heat change during neutralization

Most chemical reactions involve heat changes. Heat is always released in neutralization reactions. Therefore neutralization reactions are **exothermic reactions**. N2

CE 2011 Paper 2 Q12

Suppose we mix each pair of solutions as tabulated below in an expanded polystyrene cup and record the temperature rise.

Case	Solutions mixed	
	Strong acid	Strong alkali
1	25 cm^3 of 1 mol dm^{-3} hydrochloric acid	25 cm^3 of 1 mol dm^{-3} sodium hydroxide solution
2	25 cm^3 of 1 mol dm^{-3} nitric acid	25 cm^3 of 1 mol dm^{-3} potassium hydroxide solution

We will further discuss exothermic reactions in Topic 7 Fossil Fuels and Carbon Compounds and Topic 9 Chemical Reactions and Energy.

N2 Refer to page T30.

Video
Measuring the temperature change of the neutralization reaction between dilute hydrochloric acid and dilute sodium hydroxide solution

exothermic reaction 放熱反應

Joule (J) is a unit of energy.

We will further discuss the heat released during a neutralization reaction in Topic 9 Chemical Reactions and Energy.

The temperature rise is almost the same in both cases. For the reaction between a strong acid and a strong alkali, the heat released is 57 kJ for one mole of water produced. It is because in these neutralization reactions the only chemical change is

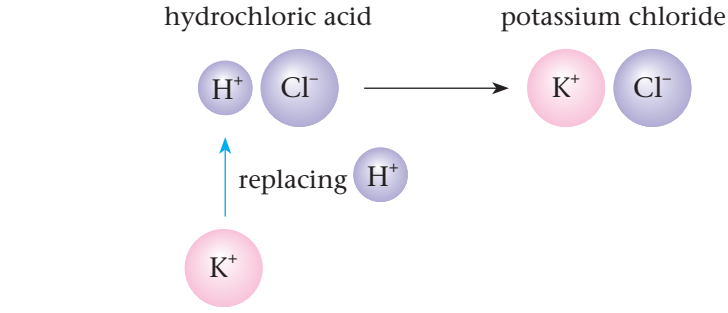
$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l) + \text{heat } (57 \text{ kJ mol}^{-1})$$

Hence the same amount of heat is released.

16.3 Formation of salts

In a neutralization reaction, salt and water are the only products. Salts form when the hydrogen ions in acids are replaced by metal ions or ammonium ions.

For example, when dilute hydrochloric acid reacts with dilute potassium hydroxide solution, the hydrogen ions in hydrochloric acid are replaced by potassium ions. The salt potassium chloride is formed.

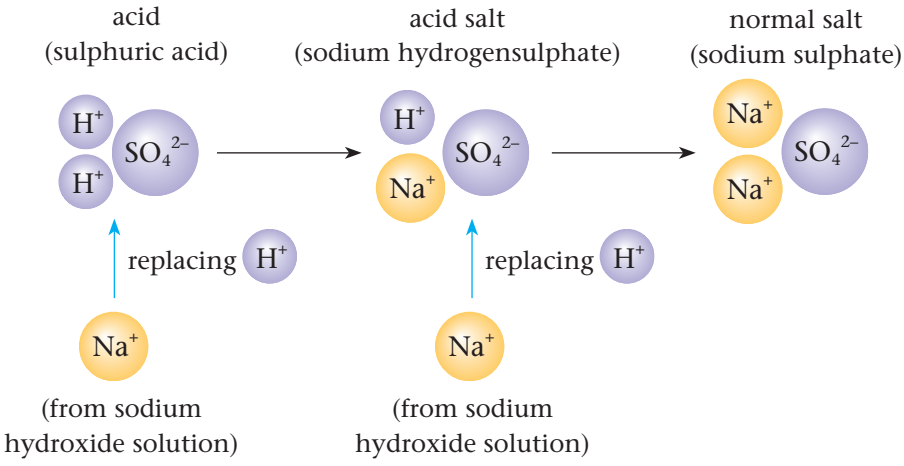


Normal salts and acid salts

The reaction of dilute hydrochloric acid with dilute sodium hydroxide solution produces only one salt, sodium chloride. This is because one molecule of hydrochloric acid can produce only one hydrogen ion.

One molecule of sulphuric acid can produce two hydrogen ions. If dilute sulphuric acid is allowed to react with dilute sodium hydroxide solution, two kinds of salt can form. When one of the hydrogen ions is replaced, the salt formed is sodium hydrogensulphate ($NaHSO_4$). It is called an **acid salt**. When both hydrogen ions are replaced, the salt formed is sodium sulphate (Na_2SO_4). It is called a **normal salt**.

joule 焦耳　acid salt 酸式鹽　normal salt 正鹽



When we mix 1 mole of sulphuric acid with 1 mole of sodium hydroxide, sodium hydrogensulphate will form.

$$H_2SO_4(aq) + NaOH(aq) \longrightarrow NaHSO_4(aq) + H_2O(l)$$

When we mix 1 mole of sulphuric acid with 2 moles of sodium hydroxide, sodium sulphate will form.

$$H_2SO_4(aq) + 2NaOH(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Most acid salts give an acidic solution when dissolved in water. For example, sodium hydrogensulphate ($NaHSO_4$) solution is acidic. This is because the hydrogensulphate ion (HSO_4^-) can dissociate to give hydrogen ion.

$$HSO_4^-(aq) \rightleftharpoons H^+(aq) + SO_4^{2-}(aq)$$

A few acid salts give alkaline solutions when dissolved in water. The explanation for the alkalinity is beyond the scope of the curriculum.

16.4 Naming of salts

A salt consists of anions and cations. The anions come from the acid and the cations come from the base.

Take potassium hydroxide solution as an example. When it reacts with hydrochloric acid, nitric acid and sulphuric acid separately, the potassium ions in the alkali will become the cations of the salts formed. Anions of the salts come from the acids accordingly.

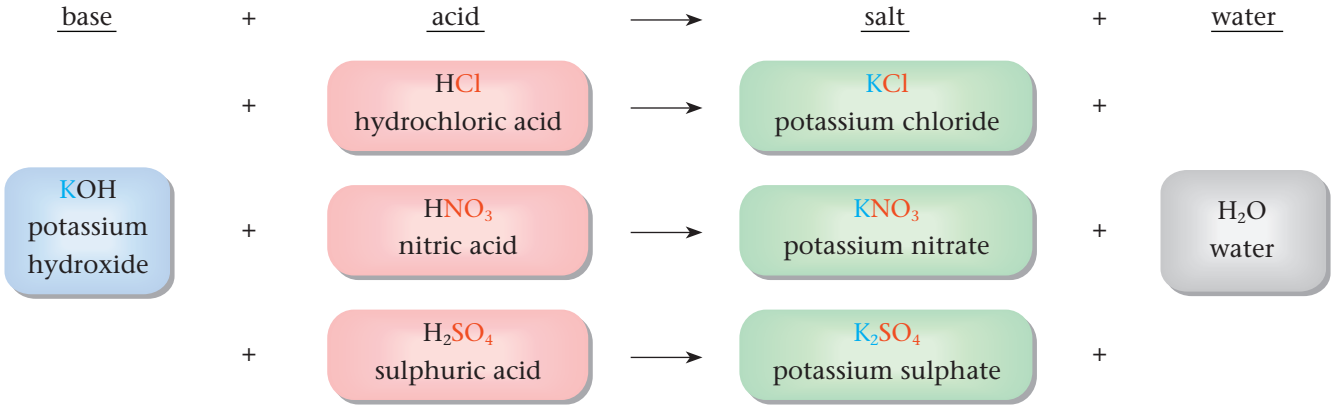


Table 16.1 lists the three most common acids and the names of the salts formed from them.

Table 16.1

Three most common acids and names of their salts

Acid	Name of salt formed when the acid reacts
Hydrochloric acid	chloride
Sulphuric acid	sulphate
Nitric acid	nitrate

The following are four helpful rules for naming salts:

1 The name of the metal or ammonium ions always comes first.

2 If the metal can form more than one type of positive ion, write a Roman numeral in brackets to show the charge of the ion.

3 If the salt contains only two elements, the name ends in -ide.

4 If the salt contains three or more elements, one of which being oxygen, the name ends in -ite or -ate.

For example, the normal salt formed from sodium hydroxide solution and sulphurous acid ($H_2SO_3(aq)$) is called sodium sulphite (Na_2SO_3). The normal salt formed from sodium hydroxide solution and sulphuric acid is called sodium sulphate (Na_2SO_4).

Table 16.2 lists the names of acid salts and normal salts formed from some common acids and bases.

Table 16.2

Names of acid salts and normal salts formed from some common acids and bases

Acid	Base	Acid salt	Normal salt
HCl	$Fe(OH)_2$	—	$FeCl_2$, iron(II) chloride
HCl	CuO		$CuCl_2$, copper(II) chloride
HNO_3	NH_3		NH_4NO_3, ammonium nitrate
CH_3COOH	NaOH		CH_3COONa, sodium ethanoate
H_2SO_4	NaOH	$NaHSO_4$, sodium hydrogensulphate	Na_2SO_4, sodium sulphate
H_2CO_3	KOH	$KHCO_3$, potassium hydrogencarbonate	K_2CO_3, potassium carbonate
H_3PO_4	NaOH	NaH_2PO_4, sodium dihydrogenphosphate Na_2HPO_4, disodium hydrogenphosphate	Na_3PO_4, sodium phosphate

Practice 16.2

Give the formula(e) and the name(s) of the salt(s) formed in each of the following reactions:

a) Aqueous ammonia reacts with dilute sulphuric acid. NH_4HSO_4 ammonium hydrogensulphate
$(NH_4)_2SO_4$ ammonium sulphate

b) Lithium oxide reacts with dilute hydrochloric acid.
LiCl lithium chloride

c) Magnesium hydroxide reacts with dilute nitric acid.
$Mg(NO_3)_2$ magnesium nitrate

16.5 Soluble and insoluble salts

To prepare a salt, we must consider the following points:

- the reactions which give the salt; and
- the reaction in which it is easy to separate the salt from the reaction mixture.

The method used to prepare a salt depends on whether the salt is soluble or insoluble in water.

CE 2009 Paper 2 Q48

N3 Refer to page T31.

Fig. 16.2 shows the solubility patterns of some salts in water. N3

Solubility patterns

Cation \ Anion	nitrate (NO_3^-)	chloride (Cl^-)	sulphate (SO_4^{2-})	carbonate (CO_3^{2-})
ammonium (NH_4^+)				
potassium (K^+)				
sodium (Na^+)				
barium (Ba^{2+})				
calcium (Ca^{2+})				
magnesium (Mg^{2+})				
zinc (Zn^{2+})				
lead(II) (Pb^{2+})				
copper(II) (Cu^{2+})				
silver (Ag^+)				

key: soluble, insoluble

CE 2009 Paper 1 Q7(a)

CE 2009 Paper 2 Q29

Fig. 16.2 The solubility patterns of some salts in water

16.6 Preparing soluble salts (except sodium, potassium and ammonium salts)

To prepare a soluble salt (except sodium, potassium and ammonium salts), mix a dilute acid with a metal, an insoluble base or an insoluble carbonate.

Video
Preparing copper(II) sulphate crystals from the neutralization reaction between dilute sulphuric acid and copper(II) oxide

- ✓ metal + acid ⟶ salt + hydrogen
- ✓ insoluble base + acid ⟶ salt + water
- ✓ insoluble carbonate + acid ⟶ salt + water + carbon dioxide

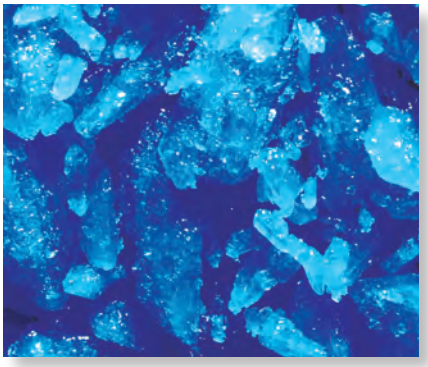


Fig. 16.3 Copper(II) sulphate crystals

Preparing copper(II) sulphate crystals

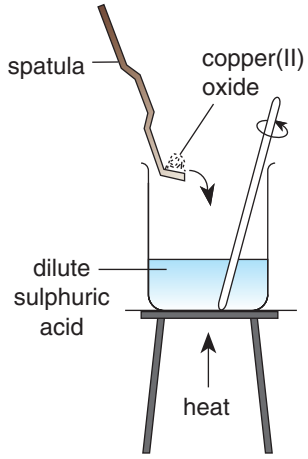
Using copper(II) oxide (an insoluble base) to react with dilute sulphuric acid forms copper(II) sulphate solution. Copper(II) sulphate (Fig. 16.3) is a soluble salt.

CE 2010 Paper 2 Q40

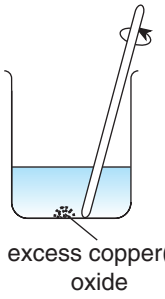
Fig. 16.4 shows the procedure for preparing copper(II) sulphate crystals by the reaction between copper(II) oxide and dilute sulphuric acid.

$$CuO(s) + H_2SO_4(aq) \longrightarrow CuSO_4(aq) + H_2O(l)$$

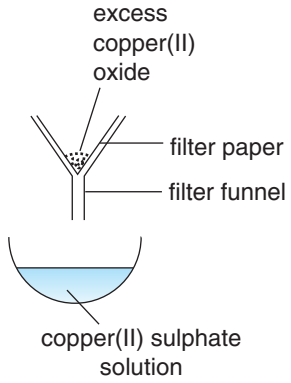
This is to ensure that the acid has reacted completely. Otherwise, the salt solution will be *contaminated* with the excess acid.



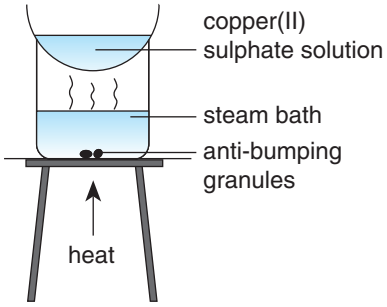
1 Add a spatula measure of copper(II) oxide into a beaker of sulphuric acid. Warm and stir the reaction mixture with a glass rod.



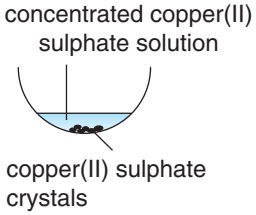
2 Add copper(II) oxide until some of it remains in the beaker.



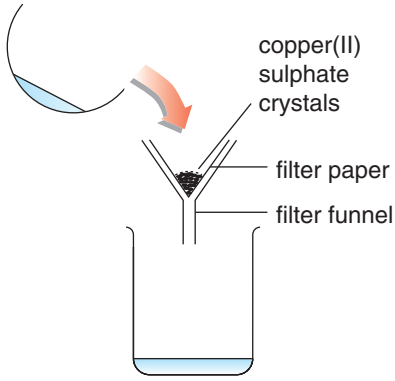
3 Remove the excess copper(II) oxide by filtration.



4 Heat the copper(II) sulphate solution to evaporate about half of the water.

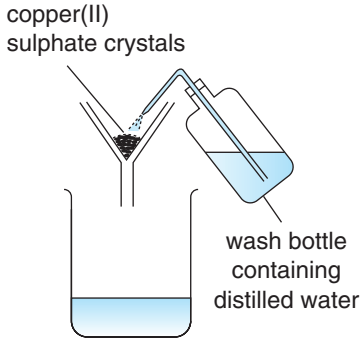


5 Set the concentrated solution aside to cool and crystallize.

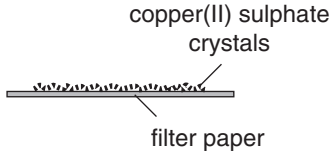


6 Filter the crystals from the remaining solution.

This is to wash off any water soluble impurities that may adhere to the crystals.



7 Wash the crystals with a small amount of cold distilled water.



8 Dry the crystals using filter paper.

Fig. 16.4 Preparation of copper(II) sulphate crystals by the reaction between dilute sulphuric acid and copper(II) oxide

contaminated 污染

16.1
Preparing magnesium sulphate crystals from the reaction between an acid and an insoluble carbonate.

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Another example is to prepare zinc sulphate by reacting dilute sulphuric acid with either zinc, zinc oxide or zinc carbonate. N4

$$Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$$

$$ZnO(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2O(l)$$

$$ZnCO_3(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2O(l) + CO_2(g)$$

N4 It is NOT appropriate to prepare zinc sulphate by mixing zinc nitrate and dilute sulphuric acid directly. This is because both zinc sulphate and zinc nitrate are soluble salts.

16.7 Preparing sodium, potassium and ammonium salts

In the above method of salt preparation, we add an excess of a solid reactant to an acid, and then filter off the unreacted solid. This ensures that all the acid is used up.

However, we cannot use the same method to prepare sodium, potassium and ammonium salts. This is because all sodium, potassium and ammonium compounds are soluble in water. Any excess solid would just dissolve. We cannot tell when the acid has completely reacted.

Solving the problem by doing a titration

We will discuss titration in detail in Unit 17.

To solve the problem, we need an acid-alkali indicator. It shows a colour change when enough alkali or carbonate is added to react with the acid completely. The process is called **titration**.

Having found out how much acid and alkali (or carbonate) are needed, we can make a pure solution of the salt by mixing those same volumes again, but without the indicator.

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

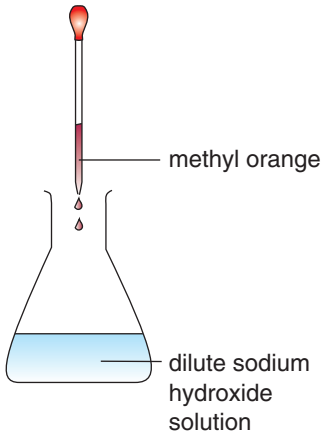
16.2
Preparing sodium chloride from an acid-alkali titration.

Preparing sodium sulphate crystals

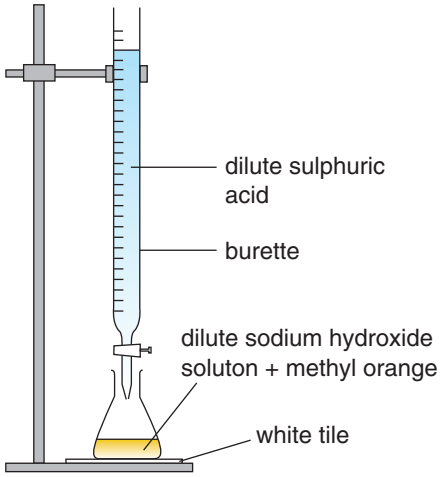
Fig. 16.5 shows the procedure for preparing sodium sulphate crystals by the reaction between dilute sulphuric acid and dilute sodium hydroxide solution.

$$2NaOH(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$$

titration 滴定法

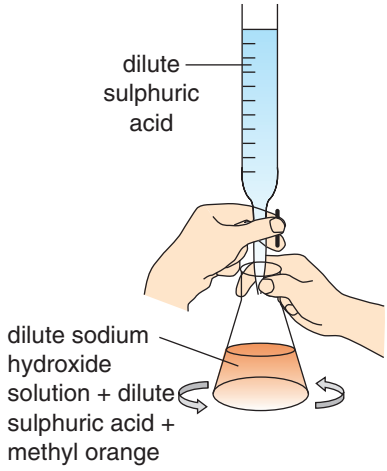


1 Place 25.0 cm^3 of dilute sodium hydroxide solution in a conical flask. Add 2 drops of methyl orange to the flask.



2 Place dilute sulphuric acid in a burette◂. Set up the apparatus as shown.

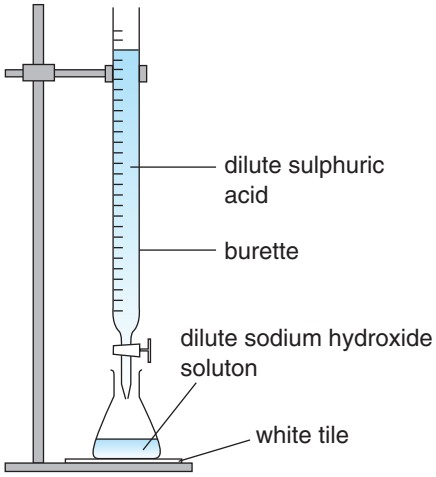
◂ We will discuss the apparatus burette in detail in Unit 17.



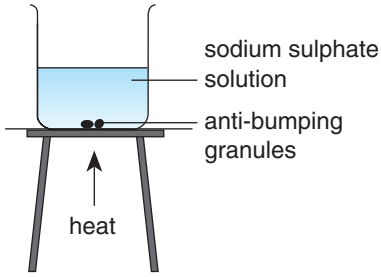
3 Run the acid from the burette until the colour of the reaction mixture just changes from yellow to orange◂. Record the volume of dilute sulphuric acid added.

◂ Notice the colour changes for methyl orange in the titration:

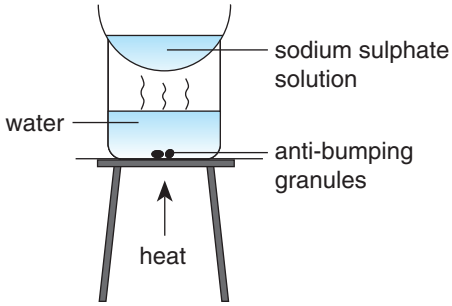
not enough acid	just right!	too much acid



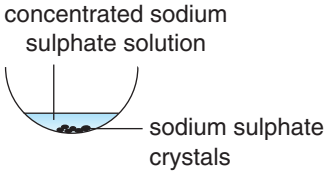
4 Mix the same volumes of acid and alkali together in a clean conical flask, *but without any methyl orange*. A sodium sulphate solution is formed.



5 Heat the sodium sulphate solution gently until $\frac{1}{3}$ of the solution is left.



6 Transfer the remaining solution to an evaporating dish. Heat using a steam bath. Evaporate about half of the water.

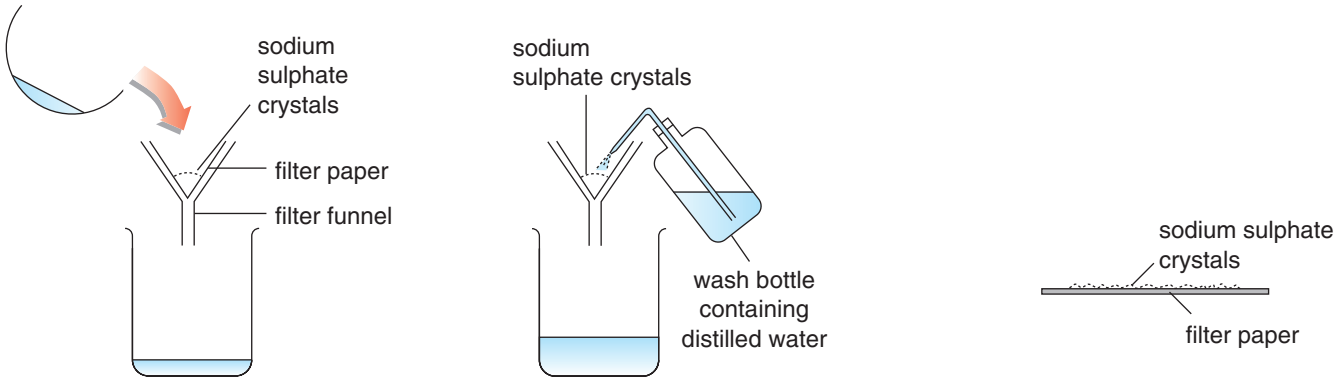


7 Set the concentrated solution aside to cool and crystallize.

Video

Preparing sodium sulphate crystals from the neutralizatrion reaction between dilute sulphuric acid and dilute sodium hydroxide solution

Continued on next page



8 Filter the crystals from the remaining solution.

9 Wash the crystals with a small amount of cold distilled water.

10 Dry the crystals using filter paper.

Fig. 16.5 Preparation of sodium sulphate crystals by the reaction between dilute sulphuric acid and dilute sodium hydroxide solution

Example 16.1

Q In a certain experiment, zinc chloride crystals are prepared from zinc according to the following steps.

Step 1 Add excess zinc to dilute hydrochloric acid.

Step 2 Remove the remaining zinc from the solution obtained.

Step 3 Evaporate the solution until it becomes saturated.

Step 4 Allow the saturated solution to cool down to obtain zinc chloride crystals.

Step 5 Separate the crystals from the saturated solution. Wash the crystals with a small amount of cold distilled water.

Step 6 Dry the crystals obtained.

a) Write the chemcial equation for the reaction between zinc and dilute hydrochloric acid.

b) For *Step 1*, explain why zinc should be in excess.

c) Suggest ONE way to show that a saturated solution has been obtained in *Step 3*.

d) For *Step 4*, explain why crystals would be obtained when the saturated solution is cooled down.

e) For *Step 5*, suggest how the crystals are separated from the saturated solution.

f) For *Step 6*, suggest how the crystals can be dried.

Continued on next page

A a) $Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$

b) To make sure that all the hydrochloric acid has reacted. / To make sure that the product is not contaminated with hydrochloric acid.

c) Remove a drop of the solution with a glass rod, and see whether any solid forms when the drop cools.

d) The solubility of zinc chloride decreases when the temperature of the solution drops.

e) By filtration

f) Absorb the water by filter paper. / Place in a desiccator.

Practice 16.3

An experiment on the preparation of sodium nitrate crystals involves the following steps:

Step 1 Transfer 25.0 cm^3 of sodium carbonate solution to a conical flask. Add a few drops of methyl orange.

Step 2 Run dilute nitric acid into the conical flask until the colour of reaction mixture turns orange. Notice the volume of acid used.

Step 3 Mix the same volumes of sodium carbonate solution and dilute nitric acid in a clean conical flask, but without any methyl orange.

Step 4 Heat the solution obtained in *Step 3* until it becomes saturated. Allow it to cool to room temperature to crystallize out the sodium nitrate crystals.

Step 5 Filter off the crystals obtained. Wash them with a small amount of cold distilled water.

Step 6 Dry the crystals.

a) Explain the purpose of adding methyl orange in *Step 1*.

b) Write the chemical equation for the reaction involved in *Step 2*.

c) Explain why a small amount of cold distilled water is used to wash the crystals in *Step 5*.

d) Suggest ONE way of drying the crystals in *Step 6*.

Practice 16.3

a) Methyl orange shows a colour change when enough dilute nitric acid is added to the sodium carbonate solution and their reaction is complete, which indicates the end point of titration.

b) $Na_2CO_3(s) + 2HNO_3(aq) \longrightarrow 2NaNO_3(aq) + H_2O(l) + CO_2(g)$

c) Washing with distilled water can remove the water soluble impurities.
Using a small amount of cold distilled water helps reduce the loss of salt.

d) Absorb the water by filter paper. / Place in a desiccator.



Fig. 16.6 Lead(II) chloride, a white precipitate, is formed when colourless solutions of lead(II) nitrate and sodium chloride are mixed

Video
Preparing lead(II) chloride by precipitation

16.8 Preparing insoluble salts

To prepare an insoluble salt, mix a solution containing cations of the salt and a solution containing anions of the salt together. When ions of the insoluble salt combine, the precipitate of the salt forms. Such a reaction is called **precipitation**. N5

For example, we can prepare an insoluble salt lead(II) chloride by mixing solutions of lead(II) nitrate and sodium chloride (Fig. 16.6). N6

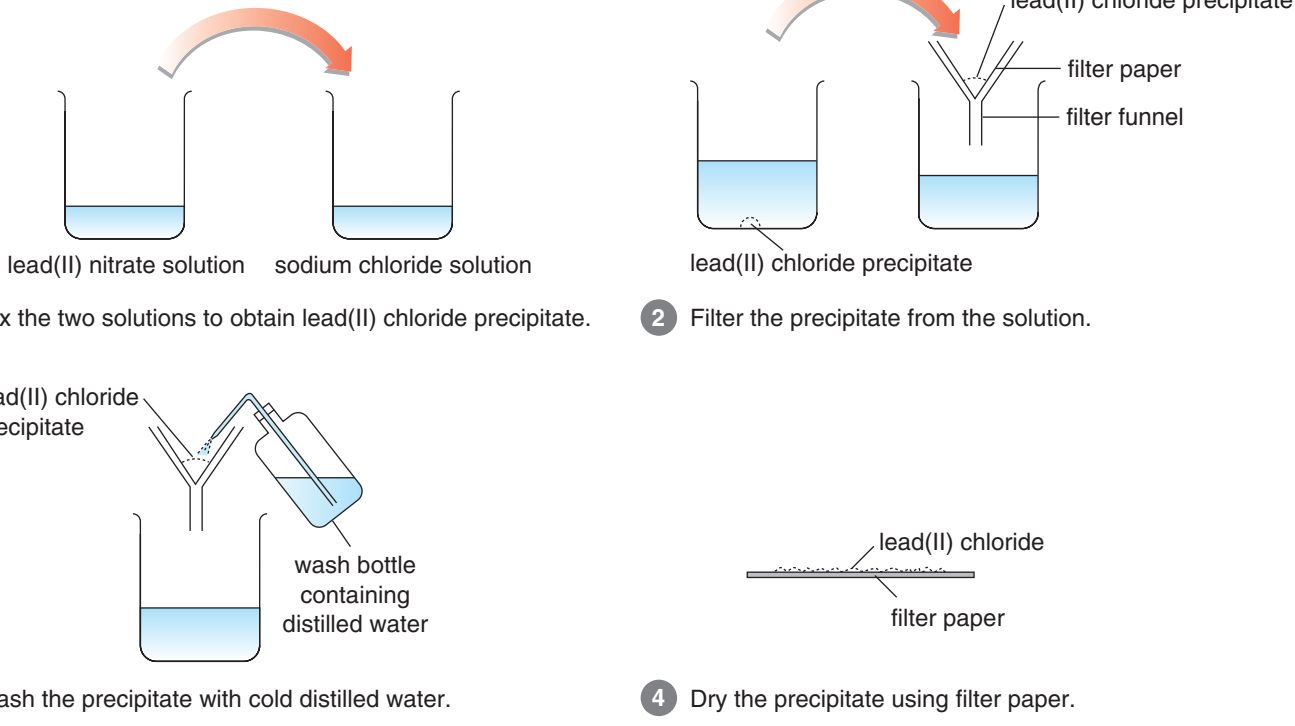
$$\underset{\text{from } Pb(NO_3)_2(aq)}{Pb^{2+}(aq)} + \underset{\text{from NaCl(aq)}}{2Cl^-(aq)} \longrightarrow \underset{\text{white precipitate}}{PbCl_2(s)}$$ N7

Fig. 16.7 shows the procedure for preparing lead(II) chloride.

The $NO_3^-(aq)$ ions from $Pb(NO_3)_2(aq)$ and $Na^+(aq)$ ions from NaCl(aq) do not take part in the reaction.

N5 Refer to page T32.

N6 Examination questions often ask about lead and its compounds. Insoluble lead(II) sulphate can be prepared by mixing lead(II) nitrate solution and dilute sulphuric acid.



1 Mix the two solutions to obtain lead(II) chloride precipitate.

2 Filter the precipitate from the solution.

3 Wash the precipitate with cold distilled water.

4 Dry the precipitate using filter paper.

Fig. 16.7 Preparation of lead(II) chloride by the reaction between lead(II) nitrate solution and sodium chloride solution

N7 Insoluble barium sulphate can be prepared from insoluble barium carbonate in the following way:

$$BaCO_3(s) \xrightarrow{HCl(aq)} BaCl_2(aq) \xrightarrow{H_2SO_4(aq)} BaSO_4(s)$$

precipitation 沉澱作用

16.3
Preparing silver chloride by precipitation.

Fig. 16.8 shows a flow diagram summarizing methods for preparing different kinds of salt.

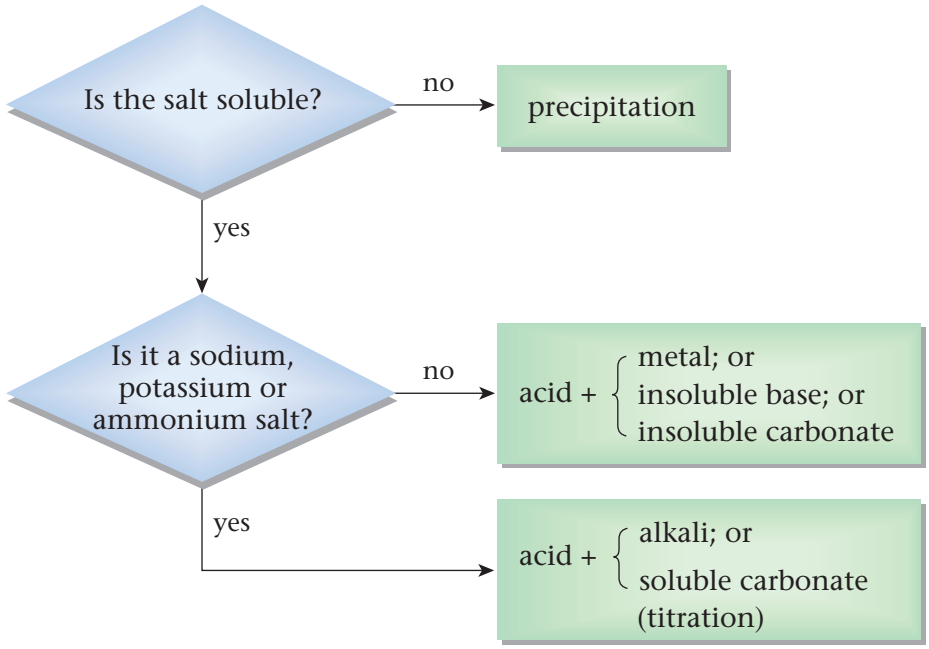


Fig. 16.8 A flow diagram summarizing methods for preparing different kinds of salt

Practice 16.4

There are three main methods for preparing salts:

A action of acid on an excess of a suitable solid

B using a titration

C using a precipitation reaction

For each of the following salts,

i) write down the letter of the appropriate preparation method;

ii) name the reactants you would use (state whether they are used as solids or solutions);

iii) write an equation for the reaction involved.

a) Barium sulphate

b) Ammonium chloride

c) Copper(II) nitrate

a) i) C
ii) Barium nitrate solution and excess dilute sulphuric acid / sodium sulphate solution / potassium sulphate solution
iii) $Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s)$
b) i) B
ii) Dilute aqueous ammonia and dilute hydrochloric acid
iii) $NH_3(aq) + HCl(aq) \longrightarrow NH_4Cl(aq)$
c) i) A
ii) Dilute nitric acid and excess copper(II) oxide
iii) $CuO(s) + 2HNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + H_2O(l)$

16.9 Uses of neutralization

Worksheet Assessment Form

A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Find & Share

Uses of neutralization

Neutralization has many useful applications, such as

- treatment of soil acidity;
- treatment of industrial waste;
- production of fertilizers; and
- use of antacids for the relief of upset stomachs.

Tasks

1 Your teacher will divide the class into groups. Each group should conduct a library search and give a 4-minute presentation on ONE application of neutralization (you may select applications other than those listed above).

Support your presentation with various aids such as diagrams, photos, etc.

2 Write a short report of no more than 500 words to summarize your findings.

Refer to page T32.

Hints for the search

1 What the problem is.

2 How neutralization is applied to solve the problem.

3 Chemical equation(s) for reaction(s) involved, if applicable.

Soil treatment

Most plants grow best when the pH of the soil is close to 7. Farmers N8 may add chemicals to the soil to adjust its pH. If the soil is too acidic, it may be treated with **quicklime** (calcium oxide) (Fig. 16.9). N9 If the soil becomes too alkaline, it may be treated with ammonium sulphate◀.

Ammonium sulphate gives an acidic solution when dissolved in water.

N8 Refer to page T33.

N9 Solid sodium hydroxide is corrosive. It is NOT appropriate to use it to neutralize acidic soil.



Fig. 16.9 A farmer is using quicklime to neutralize acidic soil

quicklime 生石灰



Fig. 16.10 Untreated liquid waste discharged from factories

N10 Using scrubbers to remove $SO_2(g)$ from flue gas in a power station is an application of neutralization.

Treatment of industrial waste

Liquid waste from factories (e.g. electroplating factories) often contains acids. The waste is harmful to water life if it is discharged into rivers and lakes without treatment (Fig. 16.10). We can add **slaked lime** (calcium hydroxide) to the waste to neutralize the acid. On the other hand, we can control the pH of alkaline waste by adding sulphuric acid to it. N10

Production of fertilizers

Many fertilizers contain chemicals such as ammonium nitrate (NH_4NO_3) (Fig. 16.11) and ammonium sulphate ($(NH_4)_2SO_4$). These chemicals are prepared by neutralization reactions.

$$HNO_3(aq) + NH_3(aq) \longrightarrow NH_4NO_3(aq)$$

$$H_2SO_4(aq) + 2NH_3(aq) \longrightarrow (NH_4)_2SO_4(aq)$$



Fig. 16.11 Ammonium nitrate pellets

Treatment of an ache caused by excess acid in the stomach

Our stomachs secrete gastric juice with hydrochloric acid. The acid helps to break down the food in our stomachs as well as to kill bacteria. Sometimes our stomachs may produce too much acid. This happens when we are under stress or have meals irregularly. We may have a burning feeling in our stomachs. The stomach ache can be relieved by taking antacids.

slaked lime 熟石灰

Weak bases are used in antacids. N11

Antacids contain bases such as magnesium hydroxide and aluminium hydroxide (Fig. 16.12). The bases neutralize the excess acid in the stomach. N12 N13

$$Mg(OH)_2(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + 2H_2O(l)$$

$$Al(OH)_3(s) + 3HCl(aq) \longrightarrow AlCl_3(aq) + 3H_2O(l)$$

CE 2009 Paper 1 Q11(a)

N11 Sodium hydrogencarbonate (commonly called baking soda) is alkaline. It is the principal antacid in most forms of Alka-Seltzer.

N12 Sodium hydroxide is NOT used as an active ingredient in antacids as it is corrosive.

N13 A suspension of magnesium hydroxide in water is sold as 'milk of magnesia'.



Fig. 16.12 A pack of antacid containing aluminium hydroxide and magnesium hydroxide

Practice 16.5

The liquid waste from a certain factory contains hydrochloric acid. The factory uses slaked lime (calcium hydroxide) to neutralize the acid in the liquid waste before discharging into the sea.

a) Why is the acid in the liquid waste neutralized before discharging into the sea?

b) Write an equation for the reaction between hydrochloric acid and slaked lime.

c) Although slaked lime is cheaper, many factories use sodium carbonate instead of slaked lime to treat their acidic liquid waste. Suggest a reason.

Practice 16.5

a) The acidity of liquid waste is harmful to water life.

b) $Ca(OH)_2(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + 2H_2O(l)$
or $Ca(OH)_2(s) + 2H^+(aq) \longrightarrow Ca^{2+}(aq) + 2H_2O(l)$

c) Calcium ion forms a number of insoluble salt. /
The liquid waste may contain sulphuric acid, which forms a layer of insoluble calcium sulphate on the surface of the slaked lime solid. Such layer hinders the reaction of slaked lime and thus decreases its efficiency in neutralization.

Have you mastered?

Key terms

neutralization 中和作用 63 exothermic reaction 放熱反應 65 joule 焦耳 66

acid salt 酸式鹽 66 normal salt 正鹽 66 chloride 氯化物 68 sulphate 硫酸鹽 68

nitrate 硝酸鹽 68 acid-alkali indicator 酸鹼指示劑 72 titration 滴定法 72

precipitation 沉澱作用 76 quicklime 生石灰 78 slaked lime 熟石灰 79 antacid 制酸劑 79

Checklist

After studying this unit, you should be able to

- ☐ write chemical and ionic equations for neutralization;
- ☐ state the water solubility of common salts;
- ☐ suggest a method for preparing a given salt;
- ☐ describe the techniques used in the preparation, separation and purification of a salt;
- ☐ name the common salts formed from the reaction of acids and bases / carbonates;
- ☐ explain some applications of neutralization.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Neutralization is the combination of hydrogen ions and hydroxide ions (or oxide ions) to form water molecules.

2 In neutralization reactions, salt and water are the only products.

acid + alkali ⟶ salt + water

acid + insoluble metal hydroxide ⟶ salt + water

acid + insoluble metal oxide ⟶ salt + water

e.g.

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

$3HNO_3(aq) + Fe(OH)_3(s) \longrightarrow Fe(NO_3)_3(aq) + 3H_2O(l)$

$H_2SO_4(aq) + CuO(s) \longrightarrow CuSO_4(aq) + H_2O(l)$

3 Neutralization reactions are exothermic reactions. Heat is released in these reactions.

4

Acid	Name of salt formed when the acid reacts
Hydrochloric acid	chloride
Sulphuric acid	sulphate
Nitric acid	nitrate

Rules for naming salts are as follows:

a) the name of the metal or ammonium ions always comes first;

b) if the metal can form more than one type of positive ion, write a Roman numeral in brackets to show the charge of the ion;

c) if the salt contains only two elements, the name ends in -ide; and

d) if the salt contains three or more elements, one of which being oxygen, the name ends in -ite or -ate.

Have you mastered?

5 The following table summarizes methods for the preparation of salts.

Salt	Method of preparation	Example(s)
Soluble salts (except sodium, potassium and ammonium salts)	acid + { metal; or insoluble base; or insoluble carbonate	Preparing zinc sulphate: $Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$ $ZnO(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2O(l)$ $ZnCO_3(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2O(l) + CO_2(g)$
Sodium, potassium and ammonium salts	acid + { alkali; or soluble carbonate (titration)	Preparing potassium chloride: $KOH(aq) + HCl(aq) \longrightarrow KCl(aq) + H_2O(l)$ $K_2CO_3(aq) + 2HCl(aq) \longrightarrow 2KCl(aq) + H_2O(l) + CO_2(g)$
Insoluble salts	precipitation	Preparing silver chloride: $Ag^+(aq)$ (from $AgNO_3(aq)$) + $Cl^-(aq)$ (from $NaCl(aq)$) $\longrightarrow AgCl(s)$

6 Uses of neutralization:

a) Acidic soil can be neutralized by adding quicklime (calcium oxide) to it.

b) Liquid waste from factories containing acids can be neutralized by adding slaked lime (calcium hydroxide) to it.

c) Fertilizers such as ammonium nitrate and ammonium sulphate are prepared by neutralization reactions.

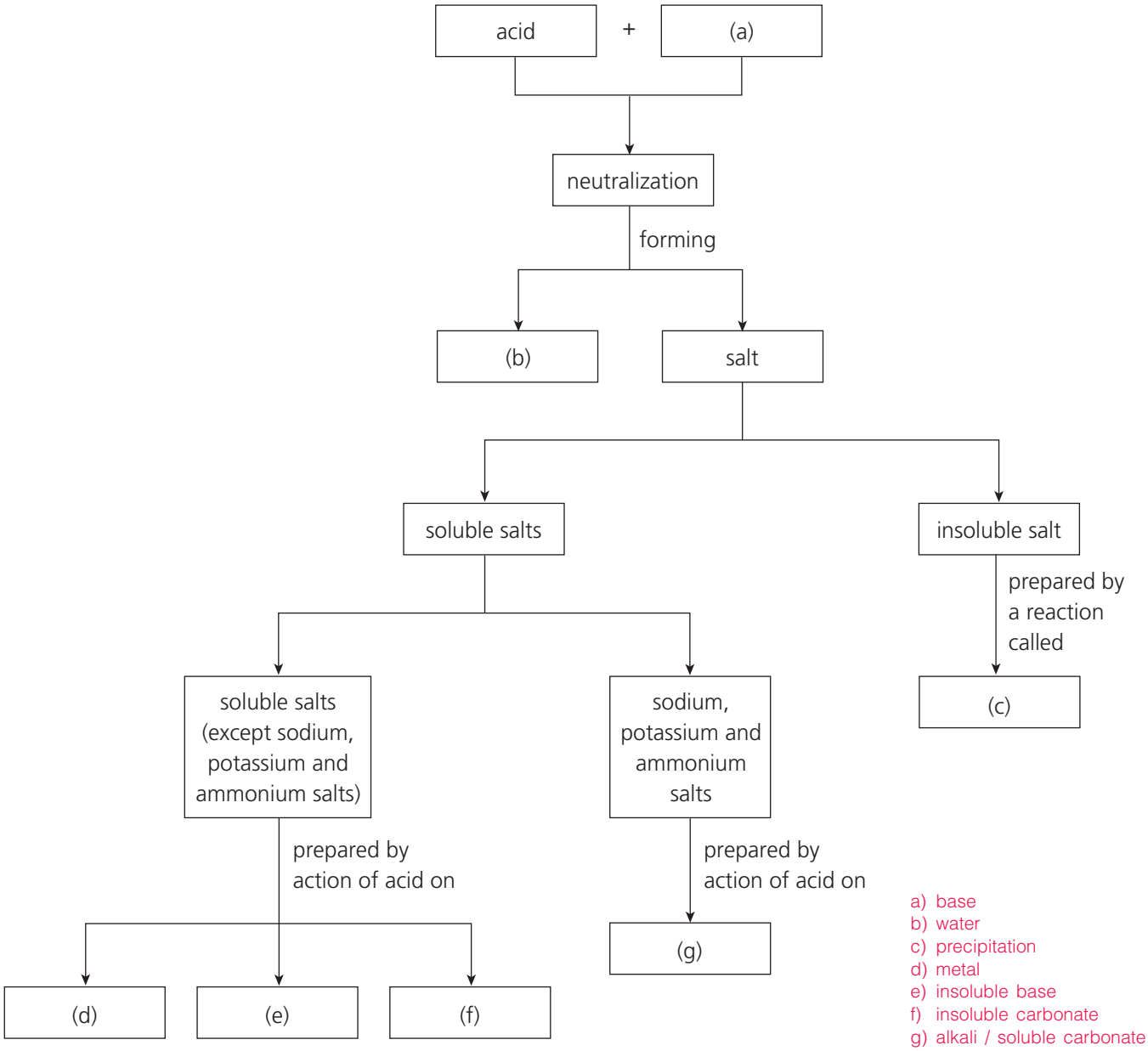
d) Antacids containing bases such as magnesium hydroxide help relieve the pain caused by excess acid in the stomach.

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.

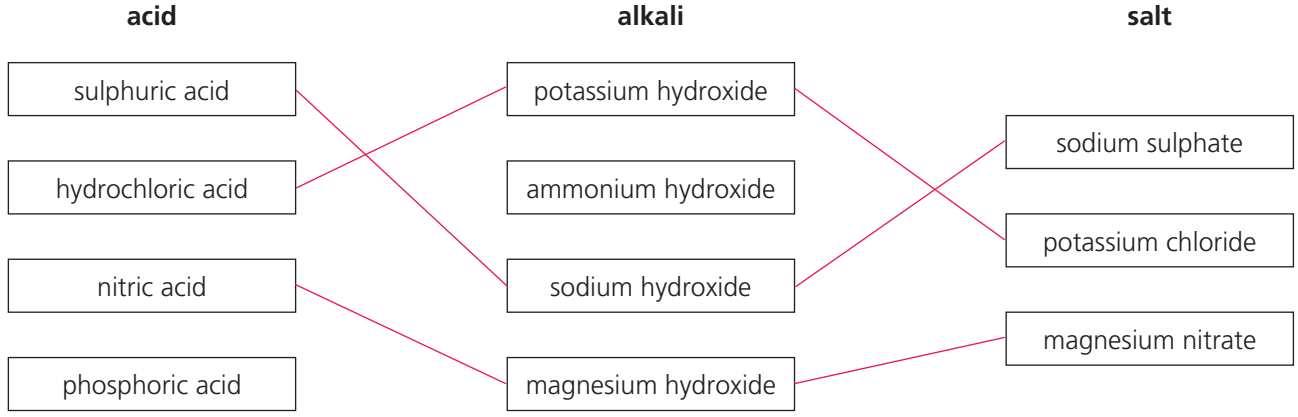


a) base
b) water
c) precipitation
d) metal
e) insoluble base
f) insoluble carbonate
g) alkali / soluble carbonate

2 An acid and an alkali react to form a salt and water.

acid + alkali ⟶ salt + water

a) Draw straight lines to join up the boxes to show which acid reacts with which alkali to make each salt.



b) i) What is the formula of the ion produced when any acid dissolves in water?
$H^+(aq)$

ii) What is the formula of the ion produced when any alkali dissolves in water?
$OH^-(aq)$

iii) Write the equation for the neutralization reaction between these two ions.
$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$

(OCR GCSE 21st Century Science (Higher Tier), Chem. A, Unit 2, Jan. 2008, 6(a)–(b))

3 Complete and balance the following equations:

a) $KOH(aq) + HNO_3(aq) \longrightarrow KNO_3(aq) + H_2O(l)$

b) $Ca(OH)_2(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + 2H_2O(l)$

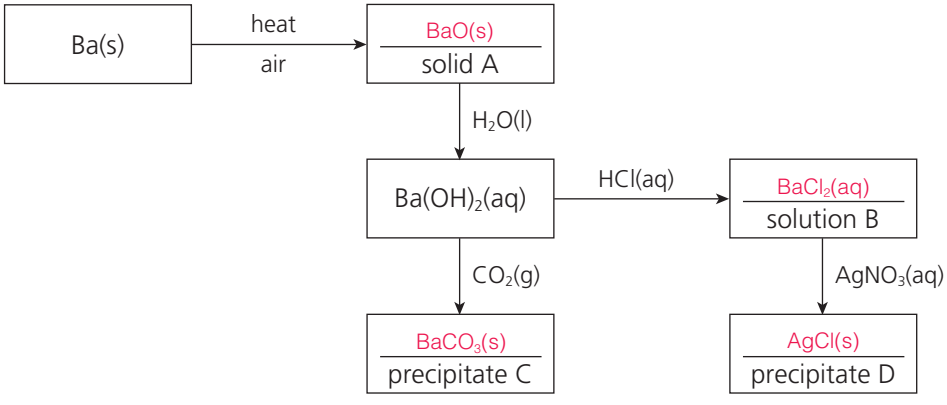
c) $2Fe(OH)_3(s) + 3H_2SO_4(aq) \longrightarrow Fe_2(SO_4)_3(aq) + 6H_2O(l)$

d) $2NaOH(aq) + H_2CO_3(aq) \longrightarrow Na_2CO_3(aq) + 2H_2O(l)$

e) $CuO(s) + H_2SO_4(aq) \longrightarrow CuSO_4(aq) + H_2O(l)$

f) $AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + NaNO_3(aq)$

4 This question looks at reactions of Group 2 elements and their compounds. The flowchart below shows some reactions involving barium. Write the formulae of substances A–D in the boxes.



(OCR Advanced Subsidiary GCE, Chemistry, 2811/01, Jun. 2009, 3(a))

Part II Multiple choice questions

5 When 25 cm^3 of 1.00 mol dm^{-3} NaOH(aq) are mixed with 25 cm^3 of 1.00 mol dm^{-3} HCl(aq), the temperature of the mixture rises by 6 °C. Which of the following reactants, when mixed under the same conditions, would give a similar temperature rise?

A 25 cm^3 of 2.00 mol dm^{-3} NaOH(aq) and 25 cm^3 of 2.00 mol dm^{-3} HCl(aq)
(B) 50 cm^3 of 1.00 mol dm^{-3} NaOH(aq) and 50 cm^3 of 1.00 mol dm^{-3} HCl(aq)
C 50 cm^3 of 0.50 mol dm^{-3} NaOH(aq) and 50 cm^3 of 0.50 mol dm^{-3} HCl(aq)
D 100 cm^3 of 0.25 mol dm^{-3} NaOH(aq) and 100 cm^3 of 0.25 mol dm^{-3} HCl(aq)

(HKDSE, Practice paper 1A, 2012, 9)

6 Which of the following pairs of reactants would react in water to give out the largest amount of heat?

A 1 mol of HCl and 1 mol of KOH
(B) 1 mol of H_2SO_4 and 2 mol of KOH
C 1 mol of $(COOH)_2$ and 2 mol of KOH
D 1 mol of CH_3COOH and 1 mol of KOH

(HKDSE, Paper 1A, 2012, 14)

7 Which row of the table shows the reagents that react with dilute hydrochloric acid to form zinc chloride?

	Zinc	Zinc oxide	Zinc carbonate
(A)	yes	yes	yes
B	yes	yes	no
C	yes	no	yes
D	no	yes	yes

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Nov. 2011, 38)

7 $Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$
$ZnO(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2O(l)$
$ZnCO_3(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2O(l) + CO_2(g)$

8 Sodium nitrate is soluble in water.

A pure sample of solid sodium nitrate is prepared in the laboratory.

Which of these methods could safely be used?

A Mix solutions of sodium chloride and potassium nitrate and filter the mixture.
B React excess sodium with dilute nitric acid and filter the mixture.
C Mix solutions of sodium chloride and potassium nitrate and obtain crystals from the solution.
(D) Neutralize sodium hydroxide solution with dilute nitric acid and obtain crystals from the solution.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Jun. 2011, 33)

Sodium salts are prepared by titration method.

9 A small amount of a powder can dissolve in water to form a clear solution. When this solution is mixed with $K_2CO_3(aq)$, a white precipitate is obtained. What can the powder be?

A Sodium sulphate
B Calcium sulphate
C Sodium hydroxide
(D) Calcium hydroxide

(HKDSE, Paper 1A, 2012, 2)

10 Which of the following reaction routes can best be used to prepare barium sulphate from barium carbonate?

A $BaCO_3(s) \xrightarrow{H_2SO_4(aq)} BaSO_4(s)$
B $BaCO_3(s) \xrightarrow{\text{conc. } H_2SO_4} BaSO_4(s)$
(C) $BaCO_3(s) \xrightarrow{HCl(aq)} BaCl_2(aq) \xrightarrow{H_2SO_4(aq)} BaSO_4(s)$
D $BaCO_3(s) \xrightarrow{\text{conc. } HCl} BaCl_2(aq) \xrightarrow{Na_2SO_4(aq)} BaSO_4(s)$

(HKDSE, Paper 1A, 2013, 8)

11 The oxide of metal Z reacts with dilute hydrochloric acid to form a colourless solution. Which of the following metals may Z be?

(1) Zinc
(2) Copper
(3) Silver

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2010, 23)

12 Which of the following reagents would form a precipitate with lead(II) nitrate solution?

(1) Dilute hydrochloric acid
(2) Magnesium sulphate solution
(3) Sodium carbonate solution

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

12 (1) $Pb(NO_3)_2(aq) + 2HCl(aq) \longrightarrow PbCl_2(s) + 2HNO_3(aq)$
(2) $Pb(NO_3)_2(aq) + MgSO_4(aq) \longrightarrow PbSO_4(s) + Mg(NO_3)_2(aq)$
(3) $Pb(NO_3)_2(aq) + Na_2CO_3(aq) \longrightarrow PbCO_3(s) + 2NaNO_3(aq)$

Part III Structured questions

Answers for the questions in Hong Kong public examinations are not provided.

13 Salts can be made by neutralizing acids.

For example, the salt magnesium sulphate is formed when magnesium, magnesium oxide, or magnesium carbonate is added to dilute sulphuric acid.

a) Complete the table to show the equations and products for these methods of making magnesium sulphate.

Method	Equation	Names of products
1	$Mg + H_2SO_4 \longrightarrow MgSO_4 +$ ___H_2___	magnesium sulphate and ___hydrogen___
2	$MgO + H_2SO_4 \longrightarrow MgSO_4 + H_2O$	magnesium sulpate and water
3	$MgCO_3 + H_2SO_4 \longrightarrow MgSO_4 +$ ___H_2O___ $+$ ___CO_2___	magnesium sulphate, ___water___ and ___carbon dioxide___

b) State ONE observation that would be made when using *Methods 1* and *3*, but NOT when using *Method 2*.
Effervescence occurs.

c) A student showed the presence of sulphate ions in magnesium sulphate solution by adding dilute hydrochloric acid and barium chloride solution.

State the observation made and name the product responsible for the observation.

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2H, Nov. 2010, 4(a)–(c))

13 c) Observation: white precipitate
Name of product: barium sulphate

14 John suffered from a stomach ache due to excess hydrochloric acid secretion. He examined the labels on two brands of antacid to find out their active ingredients. The information is shown in the following table.

Antacid	Active ingredient(s)
Milk of Magnesia	magnesium hydroxide
Alka-Seltzer	sodium hydrogencarbonate, citric acid

a) How would 'Milk of Magnesia' help John's acidity problem? Use a chemical equation to explain your answer.

b) When an 'Alka-Seltzer' tablet dissolves in water, effervescence occurs. Explain why this happens.

c) The label on 'Alka-Seltzer' indicates that the tablets should be kept in a dry place. Explain why.

14 a) Magnesium hydroxide in the milk of magnesia neutralizes the excess hydrochloric acid in the stomach and so the pain can be relieved.
$Mg(OH)_2(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + 2H_2O(l)$
b) The sodium hydrogencarbonate and citric acid in the tablet react upon dissolution in water to give carbon dioxide gas.
c) The acid will react with the sodium hydrogencarbonate in the presence of moisture.

15 When solutions are mixed together, precipitates sometimes form.

a) Barium carbonate is an insoluble compound. It is formed as a precipitate when solutions of the soluble compounds barium chloride and sodium carbonate are mixed. When solutions of the soluble compounds potassium chloride and sodium sulphate are mixed, no precipitate is formed.

Complete the table to show the results of mixing solutions of some compounds.

	Sodium carbonate solution	**Sodium sulphate solution**
Barium chloride solution	precipitate of barium carbonate	precipitate of barium sulphate
Potassium chloride solution	no precipitate	no precipitate
Calcium chloride solution	precipitate of calcium carbonate	precipitate of calcium sulphate

b) When solutions of lead(II) nitrate and potassium bromide are mixed, a precipitate of lead(II) bromide and a solution of potassium nitrate are produced.

The equation for the reaction is

$$Pb(NO_3)_2(_aq_) + 2KBr(_aq_) \longrightarrow PbBr_2(_s_) + 2KNO_3(_aq_)$$

Complete the equation by inserting the state symbols.

c) In order to prepare a pure, dry sample of lead(II) bromide, a student took the mixture produced in part (b). He then

- filtered the mixture;
- washed the solid residue with distilled water;
- left the solid in a warm place for several hours.

i) Why did the student filter the mixture?

ii) Why did he wash the solid residue?

iii) Why is it better to use distilled water rather than tap water to wash the solid residue?

iv) Why did he leave the solid in a warm place?

(Edexcel IGCSE, Chemistry, Paper 1C, May 2012, 3)

16 a) Magnesium forms the basic oxide magnesium oxide, MgO. This oxide is almost insoluble in water. On gentle warming with dilute sulphuric acid, magnesium oxide reacts to form magnesium sulphate solution.

i) Describe how you would use the above reaction to prepare a pure sample of magnesium sulphate.

ii) Suggest what action should be taken if a pupil spilt a small quantity of dilute sulphuric acid on a laboratory bench.

b) The data in the table below will be useful when answering this question.

Soluble in water	**Insoluble in water**
$MgSO_4$	$MgCO_3$ $SrCO_3$ $SrSO_4$

Magnesium carbonate reacts with dilute sulphuric acid.

$$MgCO_3(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + CO_2(g) + H_2O(l)$$

Explain why the reaction between strontium carbonate and dilute sulphuric acid stops after a few seconds.

(Edexcel Advanced Subsidiary GCE, Chemistry, Unit 1, Jun. 2009, 23(d)–(e))

15 c) i) To separate the solid and liquid in the reaction mixture, in order to obtain lead(II) bromide solid.
ii) To wash away any water soluble impurities that may adhere to the solid residue.
iii) Distilled water does not contain impurities.
iv) Any one of the following:
- To evaporate the water.
- To dry the solid.
- To increase the rate of evaporation of water.

16 a) i) Add magnesium oxide to dilute sulphuric acid until some of the oxide remain in the beaker.
Remove the excess magnesium oxide by filtration.
Heat the filtrate to evaporate about half of the water.
Set the concentrated solution aside to cool and crystallize.
Filter the crystals from the remaining solution.
Wash the crystals with a small amount of cold distilled water.
Dry using filter papers. / Put in an oven. / Put in a desiccator.
ii) Rinse with plenty of water. / Add a weak alkali (e.g. solid or aqueous sodium hydrogencarbonate).
b) Insoluble strontium sulphate forms on the strontium carbonate.

17 Insoluble salts can be made by precipitation.

The table below shows the solubility of different salts.

Soluble	Insoluble
all sodium and potassium salts	
	most carbonates
most bromides, chlorides, iodides	lead(II) and silver bromides, chlorides and iodides
all nitrates	
most sulphates	barium sulphate, calcium sulphate and lead(II) sulphate

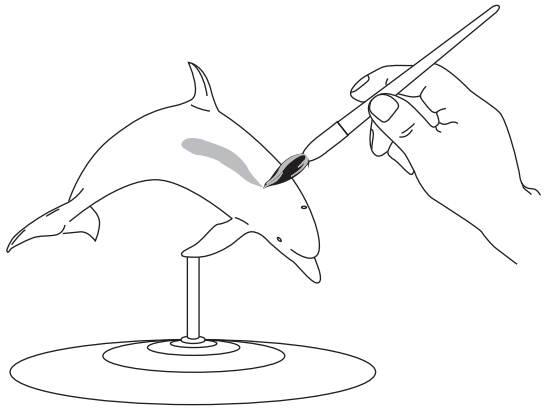
a) What does precipitation mean in the preparation of insoluble salts? The formation of a solid from the solutions of two soluble salts.

b) Which of the following salts can be made by precipitation?

Put ticks '✔' in the boxes next to the two correct answers.

- sodium sulphate ☐
- barium chloride ☐
- calcium nitrate ☐
- copper(II) carbonate ☑
- lead(II) sulphate ☑
- potassium carbonate ☐

c) Calcium sulphate is an insoluble salt which can be used to make Plaster of Paris.



Matthew wants to make calcium sulphate, $CaSO_4$. He plans to make the calcium sulphate by mixing a solution of calcium nitrate, $Ca(NO_3)_2$, with a solution of sodium sulphate, Na_2SO_4.

$$Ca(NO_3)_2(aq) + Na_2SO_4(aq) \longrightarrow CaSO_4(s) + 2NaNO_3(aq)$$

i) Solid calcium nitrate is an oxidizing agent.



Draw the hazard symbol for an oxidizing agent.

ii) Matthew dissolves 8.20 g of $Ca(NO_3)_2$ in water. He adds this solution to an excess of Na_2SO_4.

What mass of $CaSO_4$ should be produced?

(Relative atomic masses: N = 14.0, O = 16.0, S = 32.1, Ca = 40.1)

iii) Why is an excess of sodium sulphate used?

iv) Matthew filters the reaction mixture to separate the calcium sulphate from the sodium nitrate produced. He rinses the calcium sulphate with distilled water. Explain why he rinsed the calcium sulphate.

(OCR GCSE 21st Century Science (Higher Tier), Additional Applied Science A, Jun. 2009, 4)

17 c) ii) Molar mass of $Ca(NO_3)_2$ = [40.1 + 2 x (14.0 + 3 x 16.0)] g mol^{-1}
= 164.1 g mol^{-1}

Number of moles of calcium ions in 8.20 g of $Ca(NO_3)_2$ = $\frac{\text{mass}}{\text{molar mass}}$

= $\frac{8.20\text{ g}}{164.1\text{ g mol}^{-1}}$

= 5.00 x 10^{-2} mol

Molar mass of $CaSO_4$ = (40.1 + 32.1 + 4 x 16.0) g mol^{-1}
= 136.2 g mol^{-1}

Mass of $CaSO_4$ produced = 5.00 x 10^{-2} mol x 136.2 g mol^{-1}
= 6.81 g

iii) To make sure that all the calcium nitrate reacts.

iv) To wash off any soluble impurities that may adhere to the calcium sulphate.

18 Describe briefly how you would prepare each of the following salts:

Refer to page T34.

a) Iron(II) sulphate from iron

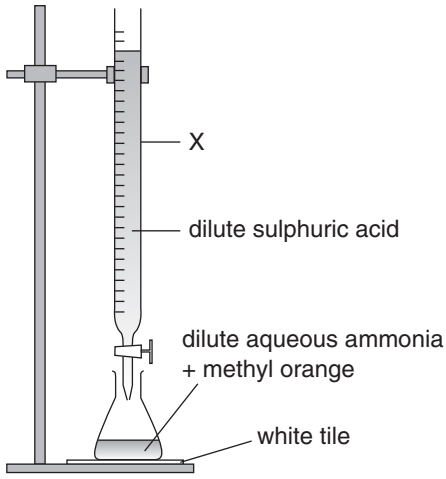
b) Copper(II) nitrate from copper(II) oxide

c) Silver chloride from potassium chloride

d) Calcium sulphate from calcium carbonate

(*Hint*: Both calcium sulphate and calcium carbonate are insoluble in water.)

19 Ammonium sulphate can be made by titrating dilute aqueous ammonia with dilute sulphuric acid.



a) Write an equation for the reaction involved.

b) 25 cm^3 of dilute aqueous ammonia were placed in a conical flask. Two drops of methyl orange were added. The set-up shown above was used for titration. 22.4 cm^3 of dilute sulphuric acid were required to neutralize the aqueous ammonia.

i) Name apparatus X.

ii) Give the colour of the dilute aqueous ammonia and indicator mixture at the start of the titration.

iii) Given the same dilute aqueous ammonia and dilute sulphuric acid, describe how you would prepare a pure, dry sample of ammonium sulphate crystals.

20 An experiment on the preparation of hydrated zinc sulphate involves the following five steps:

Step 1 Warm 30 cm^3 of dilute sulphuric acid in a beaker. Add zinc oxide to the acid until in excess.

Step 2 Filter the reaction mixture and collect the filtrate.

Step 3 Heat the filtrate until it becomes saturated. Then allow it to cool to room temperature to crystallize out hydrated zinc sulphate.

Step 4 Filter off the crystals formed, and then wash them with a little amount of cold distilled water.

Step 5 Dry the crystals.

a) For *Step 1*,

i) write the chemical equation for the reaction that occurs;

ii) suggest how one can know that zinc oxide is in excess; and

iii) explain why zinc oxide rather than sulphuric acid is used in excess.

b) Suggest ONE way to show that a saturated solution has been obtained in *Step 3*.

c) Explain why a little amount of cold distilled water is used to wash the crystals in *Step 4*.

d) Suggest ONE way of drying the crystals in *Step 5*.

e) Suggest ONE chemical that can be used to replace zinc oxide in this experiment.

(HKDSE, Practice paper 1B, 2012, 1)

19 a) $2NH_3(aq) + H_2SO_4(aq) \longrightarrow (NH_4)_2SO_4(aq)$

b) i) Burette

ii) Yellow

iii) Mix 25 cm^3 of dilute aqueous ammonia and 22.4 cm^3 of dilute sulphuric acid.
Heat the ammonium sulphate solution gently to obtain a concentrated solution.
Set the concentrated solution aside to cool and crystallize.
Filter the crystals from the remaining solution.
Wash the crystals with a small amount of cold distilled water.
Dry the crystals using filter papers. / Put in an oven. / Put in a desiccator.